Yihao Liu

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Education

Peking University Yuanpei College *Sept 2017 – Jul 2021*
Bachelor in Data Science

- **GPA**: 3.7/4.0 (Rank: 15%)
- **Advisor:** Bin Cui

Tsinghua University Institute for Interdisciplinary Information Sciences(IIS) *Sept 2021 – Now*
PH.D. in Computer Science

- **research direction**: Cloud database system / In-memory execution engine / Compression techniques.
- **Advisor:** Huanchen Zhang

Internship

Recommendation System Algorithm Engineer Intern *ByteDance, Beijing*
ByteDance *Sept 2020 – Jan 2021*

- Cold-start advertisement placement strategy exploration: proposed a strategy of "maximizing information gain" to help the system converge to an optimal strategy quickly.

Software Engineer Intern *DP Technology (start-up)*
DP Technology *Mar 2021 – Aug 2021*

- Contributor of **deepmd-kitv2.0**: improved the embedding process in the Deepmd-kit molecular simulation algorithm. https://github.com/deepmodeling/deepmd-kit

AI Research Intern *Shanghai AI Lab*
Shanghai AI Lab *May 2025 – Now*

- Diffusion Language model inference acceleration via consistency training, and KV cache optimization.

Publications

Selective Late Materialization in modern analytical databases VLDB 2025

Yihao Liu, Shaoxuan Tang, Yulong Hui, Hangrui Zhou, Huanchen Zhang
https://dl.acm.org/doi/10.14778/3749646.3749717

- The concept and trade-offs of late materialization (LM) vary greatly in modern vectorized databases compared to that in traditional column stores. Aggressively applying LM strategies no longer yields optimal results. We propose a selective LM approach that strategically applies materialization based on attributes and materialization points. After being integrated into DuckDB, this approach achieves an average latency improvement of 16% on the join order benchmark, demonstrating its effectiveness in accelerating real-world queries.

LeCo: Lightweight Compression via Learning Serial Correlations SIGMOD 2024

Yihao Liu, Xinyu Zeng, Huanchen Zhang
https://dl.acm.org/doi/10.1145/3639320

- Introduce LeCo (i.e., a Learned Compression framework), that achieves a Pareto improvement on both compression ratio and random access speed among lightweight compression techniques. LeCo enables a 5.2x speed up in the Arrow columnar execution engine and a 16% increase in RocksDB's throughput.

Cost-Intelligent Data Analytics in the Cloud CIDR 2024

Huanchen Zhang, ***Yihao Liu***, Jiaqi Yan
https://www.cidrdb.org/cidr2024/papers/p78-zhang.pdf

OkraLong: A Flexible Retrieval-Augmented Framework for Long-Text Question Answering EMNLP 2025 (Findings)

Yulong Hui, ***Yihao Liu***, Yao Lu, Huanchen Zhang
https://arxiv.org/abs/2503.02603

Rethinking Vectorized Execution: The Case for Adaptive Pipeline Fragmentation in NUMA-aware Systems Preprint

Yihao Liu*, Hangrui Zhou*,Yulong Hui,Huanchen Zhang

- This paper identifies two critical bottlenecks inherent to the long-pipeline design of vectorized databases like DuckDB: NUMA remote accesses and data chunk compaction overhead. We develop *Auto-break* to strategically break long pipelines into shorter, more efficient segments.

LATTS: LAtent space Test Time Scaling for diffusion language models Under Submission

Yihao Liu, Hangrui Zhou,Ermo Hua,Shuang cheng,Dawei Liu,Kaiyan Zhang,Huanchen Zhang,

Biqing Qi,Bowen Zhou

- Test-time scaling methods enhance auto-regressive models by extending reasoning length during inference, but they are incompatible with masked diffusion large language models. We develop LATTS, shifting the CoT process from spatial to temporal dimension, achieving performance gains using less than 12.5% additional inference-time computation: +4.2% on GSM8K, +3.4% on MATH, +3.0% on MBPP, and an average gain of 4.7% on commonsense reasoning tasks.

Interact-RAG: Reason and Interact with the Corpus, Beyond Black-Box Retrieval Under Submission

Yulong Hui, Chao Chen, Zhihang Fu, ***Yihao Liu***,Jiaping Ye,Huanchen Zhang

Nirvana: A Specialized Generalist Model With Task-Aware Memory Mechanism Under Submission

Yuhua Jiang, Shuang Cheng, ***Yihao Liu***,Ermo Hua,Dawei Liu,Che Jiang,Weigao Sun,Yu Cheng,Feifei Gao, Biqing Qi,Bowen Zhou

SDAR: A Synergistic Diffusion–AutoRegression Paradigm for Scalable Sequence Generation Preprint

Shuang Cheng,Yihan Bian,Dawei Liu,Yuhua Jiang, ***Yihao Liu***, Linfeng Zhang,Wenghai Wang,Qipeng Guo, Kai Chen,Biqing Qi,Bowen Zhou

https://github.com/JetAstra/SDAR

A Survey of Reinforcement Learning for Large Reasoning Models Preprint

Kaiyan Zhang, Yuxin Zuo, ... ***Yihao Liu***, ... , Bowen Zhou

https://huggingface.co/papers/2509.08827

Scale-Doc: Scaling LLM-based Predicates over Enormous Documents Under Submission

Hengrui Zhang*,Yulong Hui*, ***Yihao Liu***, Huanchen Zhang

https://github.com/Seurgul/ScaleDoc

G-LeCo: A GPU-Accelerated Framework for High-Throughput Learned Compression Under Submission

Youyang Xia,Feng Zhang,Jiawei Guan, ***Yihao Liu***, Huanchen Zhang,Xiaoyong Du

* means equal contribution.

Projects

Embryo: a high-performance distributed analytical database *Now*

- Main contributor to the Embryo database, developing the core logic of the execution engine and network communication between workers.

A book-sharing website with a distributed database backend *2021*

- Built a book-sharing website with a distributed NoSQL database as its backend. The database uses MongoDB, Mongos, Redis, and HDFS as building blocks.

Distributed system (MIT 6.824) *2022*

- Built a fault-tolerant distributed key-value store with linearizability.

Docking scoring function in Drug discovery *rf_vina_enhance*

- Introduced a new scoring function of docking, leveraging non-bonded interaction information and outper-

formed VinaRF20.

Technologies

Languages: C++, Python, SQL, Java

Systems worked on: RocksDB, Parquet, Arrow, Presto(Velox), DuckDB

Service

SIGMOD 2024 ARI (Availability & Reproducibility Initiative) commitee